



Bibliometric analysis of artificial intelligence cyberattack detection models

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Abstract

Cybercriminals have increasingly adopted advanced and cutting-edge methods that expand the scale and speed of their attacks in recent years. This trend coincides with the rising demand for and scarcity of highly skilled cybersecurity specialists, making them both expensive and difficult to find. Recently, researchers have demonstrated the effectiveness of Artificial Intelligence (AI) approaches in combating sophisticated cyberattacks. However, comprehensive bibliometric data illustrating the study of AI approaches in cyberattack detection remain sparse. This study addresses this gap by investigating the current state of AI-based cyberattack detection research. The study analyzed the Scopus database using bibliometric analysis on a pool of over 2,338 articles published between 2014 and 2024, including 1217 journal articles, 828 conference papers, 121 conference reviews, 85 book chapters, 70 reviews, 5 editorials, and 2 books and short surveys. The study explores various AI-based cyberattack detection approaches globally, focusing on machine learning and deep learning algorithms. The bibliometric analysis was conducted using R, an open-source statistical tool, and Biblioshiny. The findings establish that AI, particularly machine learning and deep learning, enhances intrusion detection accuracy and is a growing research trend. Researchers have effectively employed these techniques for malware detection. The USA leads in AI cyberattack research, followed by India, China, Saudi Arabia, and Australia. Despite publishing fewer articles, Canada and Italy received significant citations. Additionally, strong research collaboration exists among the USA, China, Australia, Saudi Arabia, and India. Keyword analysis highlights AI's effectiveness in identifying patterns and malicious behaviours, enhancing intrusion detection even in complex cyberattacks. Machine learning can detect intrusions based on anomalies caused by malicious or compromised devices, as well as unknown threats, with speed, accuracy, and a low false-positive rate.

Keywords Bibliometric analysis · Cybercriminals · Artificial intelligence · Cyberattack detection models

1 Introduction

The rapid integration of emerging technologies, such as edge computing, the Internet of Things (IoT), healthcare applications, and financial services, has led to significant advancements across various industries. However, this progress has also introduced critical security challenges, as cyberattacks have become increasingly sophisticated and prevalent (Erukala et al. 2022). These cyberattacks pose severe risks to these technologies, jeopardizing data integrity, privacy, and the overall security of services offered (Yu et al. 2021). Traditional cybersecurity methods, which rely heavily on human intervention and manual processes, are struggling to keep pace with the volume, complexity, and speed of modern cyber threats (Makawana et al. 2017). Consequently, the demand for more efficient, automated, and intelligent cybersecurity solutions has never been more urgent.

Artificial Intelligence (AI), particularly through machine learning (ML) and deep learning (DL) algorithms, has great potential in tackling cybersecurity challenges by enhancing the detection, classification, and mitigation of cyber threats (Makawana et al. 2017). AI-based systems can automate the identification of malicious patterns and anomalies, providing a more proactive approach to cybersecurity. However, despite significant advancements, the field of AI in cybersecurity remains fragmented. There has been limited exploration of its broader research trends and the impact of various AI models on the detection of cyberattacks.

Researchers have demonstrated the effectiveness of AI models in detecting malicious patterns. Many studies on detecting malicious threats employ deep learning and machine learning algorithms to improve the classification accuracy of known cyberattacks, detect unusual network traffic, and automate model creation (Mahbooba et al. 2021; Navya et al. 2021). Expanding on this work (Li et al. 2024), made significant progress in AI cyberattack detection by using traditional graph theory; specifically, the minimum spanning tree (MST) data structure and the medoid selection approach to enhance clustering and outlier detection techniques. This approach was designed to address the limitations of traditional MST-based clustering, particularly with datasets containing multiple densities, by proposing a scaled MST algorithm that improves clustering results and robustness. The proposed approach was validated on real-world datasets, including medical corpora, demonstrating its effectiveness and applicability and enhancing AI's ability to detect anomalies and potential cyber threats. These studies highlight the critical need for further exploratory research in this domain (Li et al. 2024). Nonetheless, they have paid little attention to bibliometric analysis. While a substantial number of studies have focused on this research domain, they primarily reflect on the impacts and trends of AI cybersecurity research.

Bibliometrics is the statistical examination of scientific and technological data, such as publications, citations, and research output (Razak et al. 2016; Firdaus et al. 2019). It enables researchers to assess their contributions to their respective domains by employing statistical and mathematical techniques to evaluate information that determines a researcher's impact in a particular field. The evaluation process includes a review of literature on scientific activities such as document types, citations, subject areas, countries, publications, publication years, collaboration, and languages. Bibliometric analysis offers numerous advantages, including the ability for researchers to critically analyze the significance of their findings and publications, foresee the future trends of their research and its societal impact, and allow institutions to examine how their publications influence subsequent research. This

research seeks to critically review the AI cyberattack detection articles published in the Scopus database between 2014 and 2024 to assess their impact, uncover global trends, and collaborations.

This study seeks to address the lack of comprehensive, systematic analysis of the AI-driven cyberattack detection research landscape. While numerous studies have focused on individual AI models and their application in detecting cyber threats, there is limited exploration of the overall trends in AI cybersecurity research, the collaborations driving this research, and the effectiveness of various AI models across different domains. Additionally, bibliometric analysis, which could provide valuable insights into global trends, research output, and the impact of AI on cybersecurity, remains largely overlooked in current literature. This study aims to fill this gap by providing a detailed bibliometric review of AI cyberattack detection models published between 2014 and 2004, exploring their impact on cybersecurity infrastructures and identifying future research directions.

To achieve the research objectives, the following research questions are posed: (1) What is the global trend of publications in AI cyberattack detection model research? (2) What insights does this trend reveal, and what are the future research directions for AI cyberattack detection models?

The scope of this study is as follows:

- i. A total of 2338 studies, including ISI-indexed articles, were obtained from the Scopus database for the bibliometric evaluation of the AI cyberattack detection model.
- ii. The experiment evaluates research efforts on AI cyberattack detection models published in different types of documents.
- iii. The dominance factor (DF) rankings were used to identify an author's dominance in the research domain. This study also focuses on the authors' citations and the keywords used in the AI-enabled cyberattack detection model research domain.
- iv. This study examines the total number of research outputs, production, and citations within each country to further evaluate AI-enabled cyberattack detection research worldwide.
- v. The keywords co-occurrence network, affiliation collaboration network, and country collaboration network are highlighted in this study. These networks aim to investigate the relationships between keywords, affiliations, and countries.

The contribution of the study is as follows:

- i. The study proposes a keyword analysis that reveals the effectiveness of AI in detecting patterns and malicious behavior, improving intrusion detection even in complex cyberattacks, and providing actionable insights for cybersecurity professionals.
- ii. The study provides insights into the global research landscape while offering an in-depth view of regional contributions.
- iii. The study uncovers evidence of robust international collaboration in advancing AI cyberattack detection, including efforts from the USA, China, Australia, Saudi Arabia, and India.

1.1 Motivation of the study

The motivation of this study is to examine the global landscape of AI cyberattack detection models and extract insights from existing research to guide future advancements in this domain. Particularly, this study investigates how AI cyberattack detection models have been utilized to enhance cyberattack detection, identify gaps in the literature, and provide actionable recommendations for cybersecurity professionals.

Furthermore, bibliometric analysis, which remains underutilized in AI cybersecurity research, serves as a powerful tool to assess and quantify the impact of this growing body of knowledge.

The importance of this study lies in the critical role AI cyberattack detection models play in strengthening cybersecurity frameworks, mitigating threats, and ensuring the integrity of digital infrastructures. Cyber threats continue to evolve in complexity, making a systematic evaluation of existing AI methodologies essential for developing more robust and adaptive security measures. Hence, this study provides deeper insights into influential research trends, key contributors, and emerging areas within AI cybersecurity models.

The benefits of this study are multifaceted, first, recognizing the most impactful researchers, institutions, nations, and journals, as well as their collaborative networks, and providing valuable insights for the scientific community. Second, mapping the temporal evolution of research themes, topics, and keywords enables researchers to identify underexplored areas and direct their investigations toward emerging trends. Lastly, this study provides important insights into future research directions, which would be of great benefit to cybersecurity professionals, policymakers, and researchers seeking to advance AI cyberattack detection methodologies.

The rest of this study is structured as follows: Sect. 2 examines related works in the research domain. The methodology used in the study is detailed in Sect. 3. Section 4 presents the study's findings. Section 5 concludes the research.

2 Related works

The study in (Makawana et al. 2017) examined bibliometric data from recent works on machine learning for cybersecurity. For researchers, educators, and authors, it offers justification and recommendations for the most recent discoveries. The findings reveal that, from January 2015 to December 2016, the most common trending outcomes from the dataset involved the detection of malicious patterns and combating adversaries.

A bibliometric analysis of cybersecurity trends using the Web of Science was presented in (Gargi and Saikat 2022). The study analyzed literature published on cybersecurity. Multiple parameters were used to analyze the data retrieved from the Web of Science database. The analysis revealed that automating cyber behavior using machine learning techniques is the current trend in cybersecurity. Another important domain is the integration of security into various components of the digital ecosystem.

The study in (Rahim 2021) presented a bibliometric analysis of cyber threats and cyberattacks in the context of higher education. The author conducted a bibliometric analysis of publications accessed from the Scopus database to assess trends and advancements in the research domain. The dynamics of the cyberattack and cyber threat landscape, as well as

the historical overview and predictions of publication trends, were also established. Similarly, the study in (Cojocaru et al. 2022) presented a bibliometric analysis of cybersecurity research articles in Eastern Europe, using Moldova as a case study. The investigation relied on data from the national database (National Bibliometric Instrument), which contains approximately 68,000 research publications, as well as two international databases (Web of Science and Scopus). The findings show that cybersecurity research is just getting started, as cybersecurity will become one of the key concerns in the information and digital-driven world.

Bibliometric analysis has also been carried out in the blockchain data privacy and security domain. For instance (Firdaus et al. 2019), presented a bibliometric analysis of blockchain. The study revealed that blockchain researchers are implementing blockchain in fields related to the data research domain, including digital storage, data privacy, big data, distributed databases, and the security of data. This demonstrates that data may be digitally secured and made freely available online with the adoption of blockchain. The adoption of blockchain to secure data in IoT and healthcare is motivated by this as well.

The study in (Dayyabu et al. 2020) presented a bibliometric analysis of cloud security. The findings indicate that the issue of data confidentiality and integrity is regarded as the most important part of cloud computing security. Another study in (Yildiz et al. 2022) presented a bibliometric analysis of cyber-physical systems and cybersecurity. The results show that the field of cyber-physical systems and cybersecurity is not yet fully developed, and more in-depth research on fundamental issues is required. Interdisciplinary research is expected to gain relevance in the coming years because this is a topic that affects many different dimensions.

The study in (Kaluri et al. 2021) examined the technological integration of the IoT-Fog-Cloud paradigm, with a focus on the security challenges present in fog-enabled IoT networks. It proposed a cooperative approach for securing edge devices through a permissioned blockchain system (pBCS), addressing key issues such as key management, communication security, and secure key exchange mechanisms. The implementation of this proposed system using private Ethereum 2.0 demonstrated effective security solutions and performance metrics. However, while the study provides valuable insights into securing fog-enabled IoT networks, it does not explore broader trends in AI-driven cyberattack detection, particularly concerning the overall cybersecurity infrastructure of IoT systems.

A bibliometric analysis was carried out on 501 research papers related to Cybersecurity and Artificial Intelligence, revealing a notable increase in publications since 2015 and indicating robust collaboration among researchers in (Albahri and AlAmoodi 2023). The study identified key application areas such as network security, deep learning, and the Internet of Things (IoT), highlighting the role of AI in enhancing Cybersecurity. The findings provide valuable insights for researchers and policymakers in addressing emerging cybersecurity challenges. However, the study lacks statistical validation to confirm the relationship between research output and impact, and it does not consider the real-time applicability.

The study in (Nakhodchi et al. 2020) presented a bibliometric analysis to examine 740 research papers on deep learning and cybersecurity, highlighting the application of deep learning algorithms in various fields such as speech recognition, image processing, bioinformatics, robotics, and cybersecurity. The study explored research trends, citation counts, and contributions from different countries, continents, research areas, authors, and institutions, demonstrating how deep learning enhances attack detection accuracy. However, it

lacks a comprehensive evaluation of the effectiveness of deep learning models in real-world cybersecurity scenarios, and does not assess the long-term impact of research across different infrastructures.

A bibliometric analysis of cybersecurity research related to neural networks was presented in (Shevchuk et al. 2024). The findings revealed a significant increase in publications from 2003 to 2023. The study explored various aspects, including publication trends, types, sources, contributing organizations, researchers, countries, and keyword patterns by employing bibliometric methods and CiteSpace software. Furthermore, timeline and burst detection analyses were conducted to identify important research trends and citation impacts. However, the study is limited to neural networks and doesn't explore other AI approaches for cybersecurity, nor does it provide a statistical validation of the research impact.

The study in (Ofusori et al. 2024) presented a literature review and bibliometric analysis focused on the application of AI in cybersecurity. The study examined 14,509 peer-reviewed research papers from the Scopus and Web of Science databases. It explored various AI techniques, including machine learning, for purposes such as anomaly detection, threat identification, predictive analytics, and automated incident response. The review provides insights into the effectiveness, challenges, and emerging trends in using AI to enhance cybersecurity. However, it is important to note that the study is limited to AI techniques and does not address the integration of AI with other advanced cybersecurity technologies.

A systematic review of 704 papers on defense mechanisms against adversarial attacks using machine learning and deep learning is presented in (Khaleel et al. 2024). The research was categorized into three main areas: security techniques, attack strategies, and defense mechanisms. The study identified key challenges such as emerging trends, and future research directions, and the need for improved mitigation methods for advanced persistent threats. However, it's important to note that the review focused exclusively on adversarial attacks and did not address other types of cyberattacks.

The study in (Arora and Jain 2021) addressed the rising concerns surrounding cybersecurity. This study identified various threats, including malware attacks, cloud abuse, and data breaches. In addition, the study identified the application areas of deep learning approaches to address these issues and identified the increasing importance of cybersecurity in today's business environment. However, the review is limited to deep learning techniques and does not consider other AI approaches.

Bibliometric analysis has been employed in various studies, with a variety of methodologies and approaches used to explore different research landscapes. A summary of relevant studies employing similar methodologies is presented in Table 1. However, there are significant differences between this study and previous works, especially regarding focus and scope. While many studies have analyzed individual AI models and their applications in cybersecurity, our work aims to provide a comprehensive, systematic analysis of the global landscape of AI cyberattack detection models. This study focuses on uncovering trends, collaborations, and the effectiveness of various AI models across different domains, while also highlighting the underutilization of bibliometric analysis in AI cybersecurity research. Furthermore, we aim to identify significant gaps in the literature and provide actionable insights to guide research collaborations and future advancements in the field of AI-driven cybersecurity.

Table 1 List of articles that adopted bibliometrics analysis method

S/N	Author(s)	Research domain	Year of publication
1	(Makawana et al. 2017)	Machine learning in cybersecurity	2017
2	(Gargi and Saikat 2022)	machine learning techniques in cyber behaviour	2021
3	(Rahim 2021)	Cyberattack in higher education	2021
4	(Cojocaru et al. 2022)	Cybersecurity	2022
5	(Firdaus et al. 2019)	Blockchain	2019
6	(Dayyabu et al. 2020)	Cloud security	2015
7	(Yildiz et al. 2022)	Cyber-physical systems and cybersecurity	2022
8	(Kaluri et al. 2021)	Security in fog-enabled IoT networks.	2023
9	(Albahri and Al-Amoodi 2023)	Cybersecurity and artificial intelligence applications	2023
10	(Nakhodchi et al. 2020)	Application of deep learning in cybersecurity	2020
11	[Erukala et al. 2022]	Neural networks toward cybersecurity	2024
12	[Fernández-Caramés 2018]	Artificial Intelligence in Cybersecurity	2024
13	Fernandez Maimo et al. [2018]	Machine learning and deep learning methods in cybersecurity	2024
14	Firdaus et al. [2019]	Deep learning in cybersecurity threats	2021

3 Methodology

This study's approach is divided into four stages: executing the search query, exporting the search results to a CSV file, findings, and analysis of findings. The search keywords for this study in the Scopus database are depicted in Table 2. This is carried out to ascertain the research direction and interest of the database in AI cyberattack detection. The Scopus database was employed for this analysis because it comprises papers that are both Scopus-indexed and ISI-ranked (Oakleaf and Oakleaf 2010). The Scopus database was used for the study because it is well-known for its thorough coverage of academic literature in a variety of areas and provides researchers with access to an extensive collection of high-quality, peer-reviewed articles and publications, which enhances the depth and breadth of the data used in the study. Reputable scientific articles are indexed by Scopus, ensuring that the selected articles meet certain quality and relevance requirements. This improves the validity and reliability of the study's findings. Additionally, restricting the study to Scopus makes the process of aggregating and analyzing data easier, promoting consistency and cohesiveness in the process. It streamlines the research workflow by enabling seamless integration with bibliometric analysis tools.

Table 2 Search phrase and keywords

Search phrase 1	(TITLE-ABS-KEY (“Artificial Intelligence Cyberattack Detection” OR “AI Cybersecurity Detection” OR “Machine Learning Cyberattack Detection Model” OR “ML Cybersecurity Detection” OR “Deep Learning Cyberattack Detection Model” OR “DL Cybersecurity Detection” OR “Cybersecurity Threat Detection using AI” OR “Advanced AI for Cyberattack Detection” OR “AI-Driven Cybersecurity Detection” OR “Intelligent Cyberattack Detection Models”)) AND PUBYEAR>2014 AND PUBYEAR<2024
Search phrase 2	(TITLE-ABS-KEY (“AI Cybersecurity Detection” OR “Machine Learning Cyberattack Detection Model” OR “ML Cybersecurity Detection” OR “Deep Learning Cyberattack Detection Model” OR “DL Cybersecurity Detection” OR “Cybersecurity Threat Detection using AI” OR “Advanced AI for Cyberattack Detection” OR “AI-Driven Cybersecurity Detection” OR “Intelligent Cyberattack Detection Models”)) AND PUBYEAR>2014 AND PUBYEAR<2024

The next phase involves implementing the search query. The search query returned a total of 2338 articles published between 2014 and 2024. This study focused on current trends and articles published within the last five years; hence, this analysis is limited to articles published between 2014 and 2024. The selected article’s time frame (2014 to 2024) aligns with the objective of the study, which involves analyzing current trends and developments in AI cyberattack detection. This time frame enables a thorough analysis of current developments, ensuring that the data is current, applicable, and relevant.

The bibliometric analysis was developed using the R programming language, an open-source statistical tool. We specifically installed and used the bibliometrix library in the R desktop software package (Aria et al. 2017). This bibliometrix tool has been used in numerous studies. However, this is the first study to use this tool to perform bibliometric analysis in AI cyberattack detection model research. The tool provides a variety of AI cyberattack detection model insights, which we visualized and investigated in subsequent sections.

The study then focused on exploring knowledge and other features of the research domain, such as examining various research domains, document types, authors, citations, and countries. The final stage involves implementing the bibliometric analysis and retrieving the findings. The AI-enabled cyberattack detection model analysis was built using the R open-source statistical tool. The bibliometrix package, in particular, was installed and used to carry out the bibliometric analysis.

The study employed the Dominance Factor (DF) ranking method to assess the authors’ prominence in cyberattack detection research output. DF represents the ratio of multi-authored articles in which an author is listed as the first author. It is calculated by dividing the number of multi-authored articles where the author is the first author (Nmf) by the total number of multi-authored papers (Nmt) authored by the individual. In cases of single-author articles, this calculation is excluded, as the value is constantly 1.

The model’s practicality is demonstrated by its ability to perform real-time data analysis efficiently. This involves processing data as soon as it becomes available and ensuring that insights and trends remain current and relevant. Additionally, computational efficiency is essential, as the use of tools like R and the bibliometrix package prioritizes algorithms and processes that handle large datasets efficiently. As a result, processing delays are minimized.

The proposed model's ability to visualize and investigate insights from the analysis is also important for real-time application. This ensures the generation of visualizations and interpretations in real-time, which are essential for informed decision-making and guiding future research directions.

The study implemented automated data processing, analysis, and visualization. This approach enhances cost efficiency by reducing manual labour costs. Additionally, the use of open-source tools like R and the bibliometrix packages improves cost efficiency by reducing software licensing fees. The steps and procedures employed in this investigation are depicted in Fig. 1.

The study employed the chi-square test to analyze the research output and its impact on AI cyberattack detection. A null hypothesis was formulated to assess the validity of the model.

- i. Null Hypothesis (H_0): There is no significant correlation between the number of articles published by a country and the total citations received by those articles in AI cyberattack detection research.
- ii. Alternative Hypothesis (H_1): There exists a significant correlation between the number of articles published by a country and the total citations received by those articles within AI cyberattack detection research.

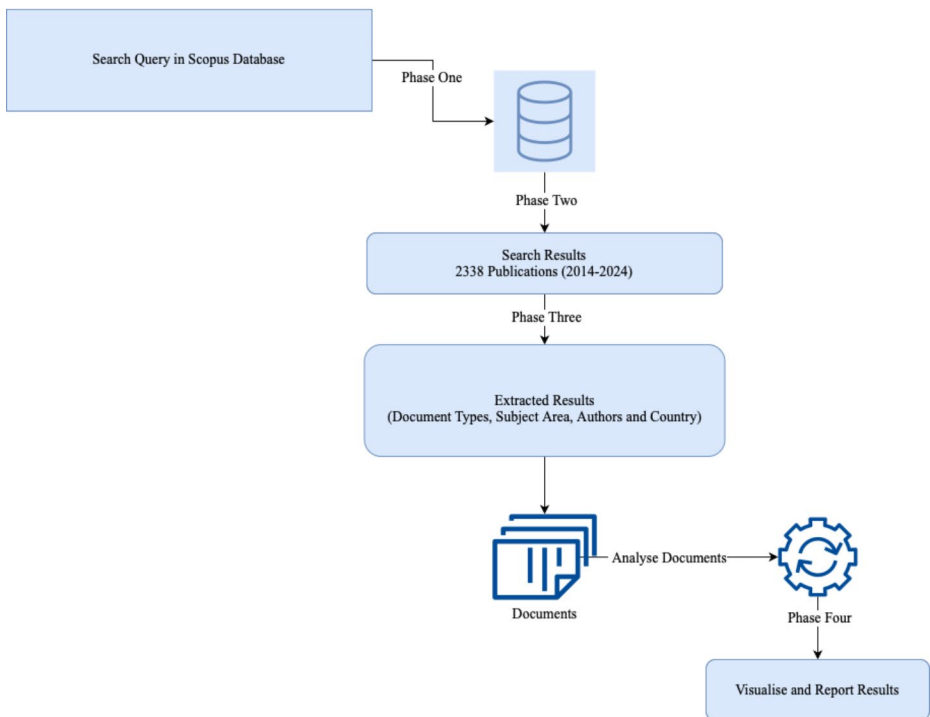


Fig. 1 Phases of the research methodology

3.1 Screening and eligibility criteria

A two-step procedure outlined in Fig. 2; Table 3 was used to screen the articles retrieved during the search procedure and exported to the Excel CSV spreadsheet by two of the authors. Only article titles and abstracts were screened in the first phase. The relevant data were exported to an Excel CSV spreadsheet for the screening process, which included the author names, article titles, DOIs, publication years, and article types. The second phase involved reading the entire publication. Articles deemed irrelevant for the analysis were excluded. Finally, bibliometric analysis was applied to the articles that met the eligibility criteria and passed the second screening phase.

The initial screening involved reviewing titles and abstracts to filter out articles from scientific domains other than those related to the research domain. In the screening process, the following exclusion criteria were applied:

- i. Articles that are written in more than one language, except the title and abstract, which are reported in English. These articles could not have been excluded previously using the Scopus filter options.



Fig. 2 Screening of the articles

Table 3 Screening and eligibility criteria

Element	Exclusion criteria	Inclusion criteria	Screening phase
Research domain	Articles not related to AI cybersecurity detection approaches	Articles that employed AI approaches to detect cyberattacks.	Both
Research outcome	lack of a definition for the search phrases listed in Table 2. Articles in AI that do not particularly discuss AI-driven cybersecurity, or attack detection approaches, and current trends or AI techniques in cyberattack detection.	Articles that include the definition of the search phrases listed in Table 2. Articles such as systematic literature reviews and peer-reviewed mappings that address cybersecurity issues using machine learning or deep learning techniques.	Both
Language	Articles that are written in more than one language, with the exception of the title and abstract, which are reported in English	Articles written in English language.	Both
Type of publication	Thesis, project reviews and website articles	Peer-reviewed articles, review papers, conference proceedings, book chapter, short survey, and editorial.	Both

- ii. Lack of a definition for the search phrases listed in Table 2.
- iii. Articles not related to AI cybersecurity detection approaches.
- iv. Articles in AI that do not particularly discuss AI-driven cybersecurity, attack detection approaches, or current trends or AI techniques in cyberattack detection.

The following inclusion criteria were applied to select the relevant articles for the study:

- i. articles that employed AI approaches to detect cyberattacks.
- ii. articles written in English language.
- iii. articles such as systematic literature reviews and peer-reviewed mappings that address cybersecurity issues using machine learning or deep learning techniques.
- iv. articles that include the definition of the search phrases listed in Table 2.

4 Bibliometric analysis

The bibliometric analysis is categorized into four components and numerous subcategories. The primary categories include document types, research domains, countries, and authors. The author category was divided into author keywords, author citations, the total number of citations, country of publication, country's total number of publications, country's total

Table 4 Extracted data main information

Description	Results
Sources (Journals, Conferences, Books, etc.)	897
Keywords Plus (ID)	8408
Author's Keywords (DE)	4328
Authors	6298
Single-authored documents	249
Multi-authored documents	6183

Table 5 Document distribution

Document type	Number of documents
Conference Paper	1217
Article	828
Conference Review	121
Book Chapter	85
Review	70
Book	10
Editorial	5
Short Survey	2

number of citations, and the collaboration networks. These findings are important because they provide bibliometric data that can be used to identify high-impact research and unearth new data for AI-enabled cyberattack detection research.

4.1 Extracted document

Table 1 illustrates the information that was extracted from the Scopus database and pertains to 2338 publications published between 2014 and 2024. These articles were written from 94 sources, consisting of conference papers, proceedings, books, book chapters, and journals. A total of 4328 authors' keywords and 8,408 keywords plus were identified in the retrieved articles. A total of 6298 authors were identified from the retrieved articles, with 249 single authors and 6,183 multiple authors. From 2014 to 2024, the results showed that the number of publications increased exponentially. This figure is expected to rise even further by the end of 2022 and 2023. The examination of the various types of extracted documents is shown in Fig. 2. The results show that conference papers had the most documents, with a total of 1217, followed by journal articles with 828 and conference reviews with 121. Tables 4 and 5 show the extracted data's primary information as well as the distribution of the documents.

Figure 3 depicts a trend in which AI-enabled cyberattack detection researchers are increasingly interested in publishing their studies in conference proceedings instead of accumulating studies published in journals or conference reviews. This is because these conference papers were submitted before the conference, and all participants now have access to them. This situation enables participants to understand the papers' core concepts and, where necessary, provide feedback after the papers have been presented and the conference has ended. Based on the feedback, the authors might decide to revise and improve their original methodology. Since readers can read conference publications in advance of the conference more than once and understand the findings, they are significantly more helpful.

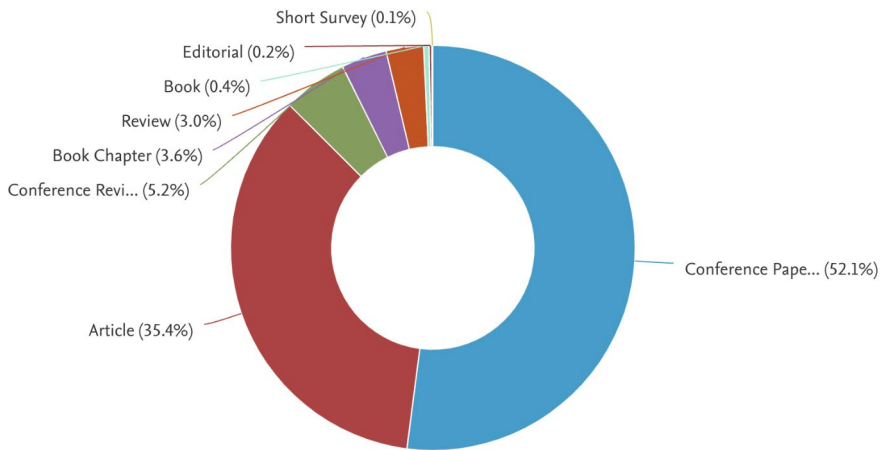


Fig. 3 Document distribution

As a result, the authors in the research domain are much better able to gather participant feedback during the conference.

Table 6; Fig. 4 demonstrate that IEEE Access is the first outlet that researchers working on AI-enabled cybersecurity detection choose to publish in. Many concepts and novel contributions that used methods for AI-enabled cyberattack detection were contained in IEEE Access. Figure 4 suggests that in the future, IEEE Access will publish an increasing number of articles on AI-enabled cybersecurity detection models. In addition to IEEE Access, Fig. 4 depicts the frequency of occurrences of all sources, including lecture notes, ACM, Advances in Intelligent Systems and Computing, and others.

4.2 Distribution of publications

Table 7 depicts the number of papers published between 2014 and 2024. Throughout this period, the number of publications has steadily increased, though at a slower rate in 2023. Table 7 shows that 2022 had the highest number of publications (792), followed by 2021 (722), while 2023 and 2014 had the least publications, respectively. The reason for this is that this research domain started in 2014. Additionally, we have yet to enter the first quarter of 2023. Hence, we expect additional articles in this research domain to be published before the end of 2023.

4.3 Subject area

The subject area that involves AI-enabled cyberattack detection research is shown in Fig. 5. As was to be predicted, computer science was the field that attracted the most interest from researchers, with 42.5% involvement in AI-enabled cyberattack detection models. Additionally, the development of the AI-enabled cyberattack detection model was sparked

Table 6 Top 20 most relevant sources

Source	Number of documents
IEEE access	113
Lecture notes in computer science including subseries lecture notes in artificial intelligence and lecture notes in bioinformatics	100
ACM International Conference Proceeding Series	68
Communications In Computer And Information Science	56
Advances In Intelligent Systems And Computing	50
Sensors	50
Lecture Notes In Networks And Systems	43
Computers Materials And Continua	33
Applied Sciences Switzerland	30
Ceur Workshop Proceedings	30
Electronics Switzerland	30
Computers And Security	29
International Journal Of Advanced Computer Science And Applications	28
Lecture Notes On Data Engineering And Communications Technologies	26
Proceedings Of SPIE The International Society For Optical Engineering	19
IEEE Internet Of Things Journal	18
Computers And Electrical Engineering	17
Studies In Computational Intelligence	15
Future Generation Computer Systems	14
IEEE Transactions On Industrial Informatics	13

by the union of the two fundamental fields of computer science: machine learning and cybersecurity.

Machine learning can help discover new anomalies by developing robust models from the fed data. Machine learning is very efficient in identifying patterns and extracting malicious vectors from massive volumes of data where humans are ineffective (Pupillo et al. 2022; Alshehri 2022). Asynchronous profiling, evaluating variation from behavioral patterns, is difficult to execute on massive datasets. However, machine learning can perform efficient behavioral analysis on a large volume of the dataset to detect malicious patterns (Pupillo et al. 2022). A machine learning model based on persistent attackers' Tactics, Techniques, and Procedures (TTP) behavior (as described in the Mire attack scheme) may help in long-term and wide-reaching attacker identification (Pupillo et al. 2022). Based on an analysis of user behaviors, the machine learning approach is considered the most suitable candidate to detect cyberattacks. These advantages of machine learning led machine learning approaches to be one of the major players in cyberattack detection. Engineering, mathematics, and decision sciences are the next three areas of interest, followed by physics and astronomy, and materials science. Mathematics and engineering were always going to be the most involved subjects since cybersecurity requires rigorous mathematics and engineering approaches that can be used to handle security challenges, evaluate security controls, and investigate security breaches. It is worth noting that AI methodologies such as machine learning and deep learning cybersecurity models have built-in mathematical prerequisites.

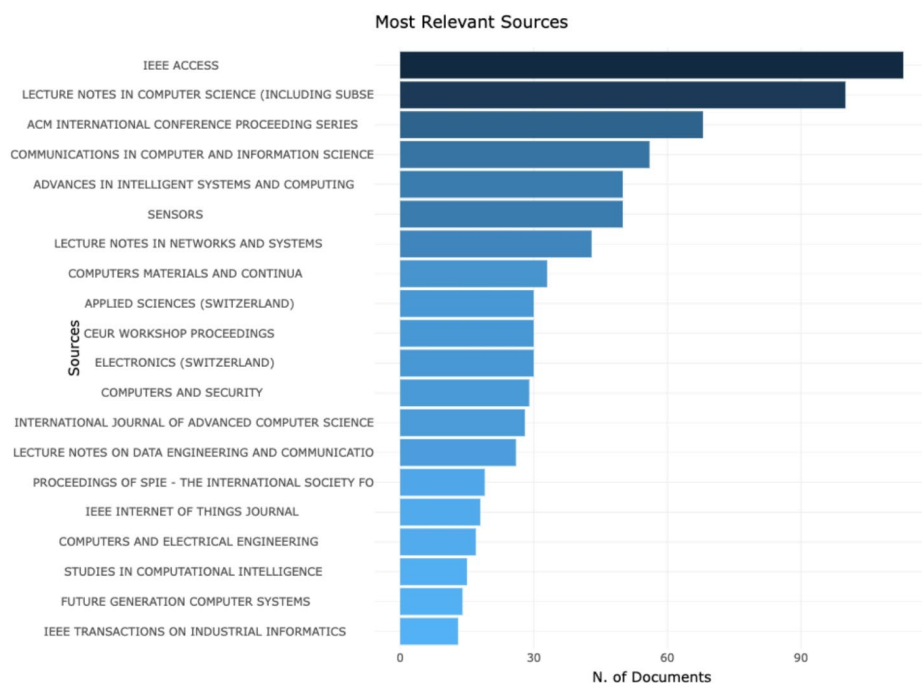


Fig. 4 Top 20 most relevant sources

Table 7 Distribution of publica-
tion based on year

Year	Number of Publications
2023	13
2022	792
2021	722
2020	360
2019	230
2018	130
2017	49
2016	26
2015	11
2014	5

These AI cybersecurity models are powered by a combination of calculus, probability, linear algebra, and statistics.

It is worth noting that researchers are paying close attention to the health and medical fields, which are currently incorporating AI-enabled cybersecurity models into their cybersecurity architecture. This is because COVID-19 makes the healthcare system more vulnerable to cyberattacks (Arpita et al. 2022). There are frontline workers and sensitive data in this critical situation; hence, it is important to maintain a high level of security in the healthcare sector. Researchers believe that an AI-enabled cybersecurity model can enhance the security of medical data and efficiently detect malicious patterns in hospital cyberspace

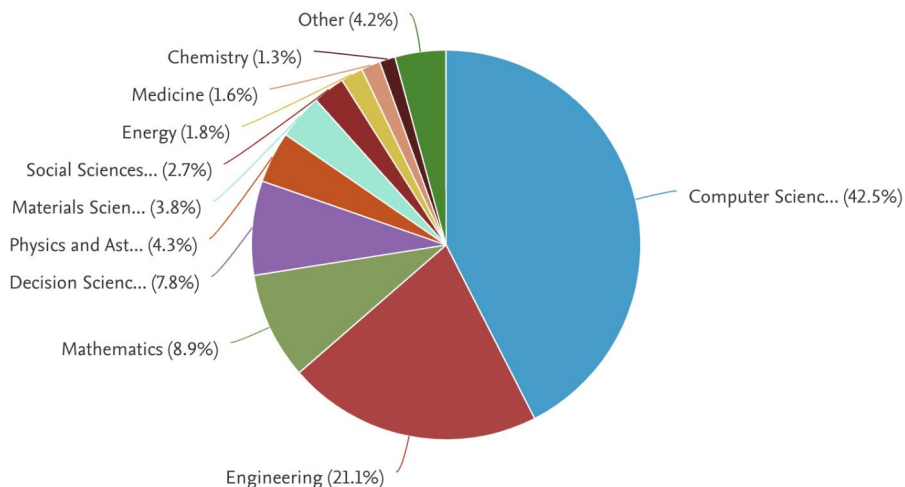


Fig. 5 Subject area distribution

(Arpita et al. 2022). The subject areas related to health and medicine include medicine and chemistry.

4.4 Authors

The authors who participated in AI-enabled cybersecurity research the most are listed in this section. This section used the author's keywords, the dominance ranking factor, and the total number of citations to achieve this. Table 8; Fig. 6 include the authors whose articles have been published and are among the top 20.

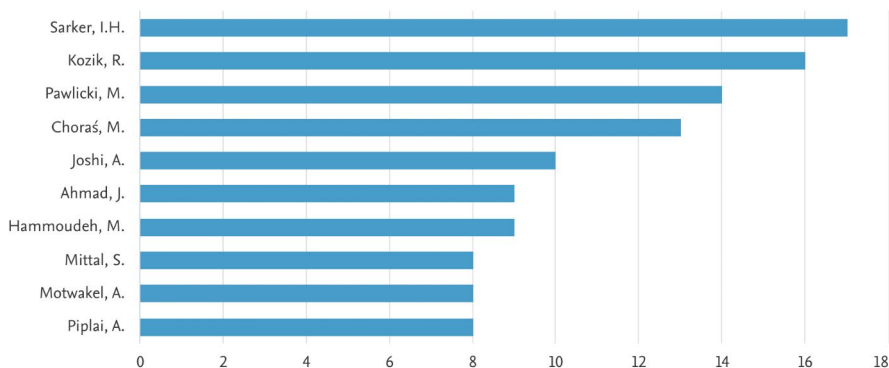
Table 8; Fig. 6 show that Sarker leads with 17 publications, followed by Kozik (16), Pawlicki (14), Choras (13), and Joshi (14). The top five authors have fifteen, fourteen, twelve, and ten articles, respectively. The remaining authors were found to have published nine, eight, seven, six, five, four, or three articles, respectively. The results in Table 8 illustrate that these authors contributed to the publications as either the primary author or the corresponding author. It should be noted that some authors only published as the primary author, while others published with co-authors. As a result, it is necessary to evaluate each author's level of contribution. This is accomplished by looking into the dominance ranking factor in the next section through several factors.

4.5 Dominance ranking of authors

The dominance factor (DF) is a ratio that determines the percentage of multi-authored articles in which one author is the lead author (Kumar and Kumar 2008). The DF ranking assesses an author's uniqueness in article publication. The DF factor is determined by dividing the total number of multi-authored articles with the author (Nmt) by the number of multi-authored publications with the author as the first author (Nmf). This is omitted in the

Table 8 Top 20 authors with numbers of publication

Author(s)	Number of publications
Sarker, I.H.	17
Kozik, R.	16
Pawlicki, M.	14
Choraś, M.	13
Joshi, A.	10
Ahmad, J.	9
Hammoudeh, M.	9
Mittal, S.	8
Motwakel, A.	8
Piplai, A.	8
Sarigiannidis, P.	8
Shahriar, H.	8
Alazab, M.	7
Chen, H.	7
Ochiai, H.	7
Ozawa, S.	7
Alotaibi, S.S.	6
Alqahtani, H.	6
Anwar, A.	6
Ban, T.	6

**Fig. 6** Top 20 authors with numbers of publication

case of a single author because single-authored publications always have the value “one.” Eq. 1 depicts the mathematical equation for DF.

$$DF = \frac{Nmf}{Nmt} \quad (1)$$

Table 9 depicts the top 20 DF rankings. The result indicates that the ranking is led by Sarker, who contributed as the first author to six articles and as a co-author to seven articles. Table 9 shows the DF, the importance of first authorship, multi-authorship, and rank by articles. Zhao R.Y. and Dastidar P.G. are the top two players in the DF ranking. A more thorough

investigation reveals that Pass Rafael is a professor at Swinburne University of Technology, focusing on machine learning, data science, AI-driven cybersecurity, and threat intelligence. Kozik is a cybersecurity researcher at the University of Technology and Life Sciences, who has supervised many students and published multiple articles on machine learning cybersecurity approaches. These results show that they share a common interest in the AI-enabled cybersecurity model, which significantly influences their contributions to the research domain.

The next section explores the keywords that the authors employed in their publications, including the specific approaches adopted and the features or subject areas presented in the keywords related to the AI-enabled cybersecurity model.

4.6 Keywords of retrieved articles

This section provides an overview of the relationship between AI-enabled cyberattack detection approaches and keywords. The authors included multiple keywords related to AI-enabled cyberattack detection in their articles. This is a significant step as it helps analyze current research trends, identify knowledge gaps in the field of AI-enabled cyberattack detection research, and highlight other areas of study where AI-enabled cyberattack detection is being explored.

Table 10 illustrates the top 20 authors' keywords. The ranking is led by "cybersecurity," followed by "machine learning," "network security," "cybersecurity," "intrusion detection," and "deep learning." The fundamental concept of cybersecurity is to protect resources in

Table 9 Top 20 DF ranking

DF ranking	Authors	DF	Multiple authored	First authored	Articles ranking
1	Zhao RY	1.0000000	3	3	6
1	DastidarPG	1.0000000	2	2	10
1	Milman BL	1.0000000	1	1	10
1	Schwechheimer H	1.0000000	2	2	10
1	Shiau WL	1.0000000	2	2	10
1	Tseng YH	1.0000000	2	2	10
7	Small H	0.6666667	3	2	1
7	Ahlgren P	0.6666667	3	2	6
7	Tsay MY	0.6666667	3	2	6
10	Zitt M	0.6000000	5	3	2
10	Glanzel W	0.6000000	5	3	3
10	Huang MH	0.6000000	5	3	3
13	Thijs B	0.5000000	4	2	5
13	Boyack KW	0.5000000	2	1	10
13	Chen CM	0.5000000	2	1	10
13	Klavans R	0.5000000	2	1	10
13	Meyer M	0.5000000	2	1	10
13	Ni CQ	0.5000000	2	1	10
13	Song M	0.5000000	2	1	10
20	Qiu JP	0.3333333	3	1	6

cyberspace from unauthorized users. Cybersecurity is also considered the application of processes, technologies, and controls to protect network infrastructures, computing systems, devices, data, and programs from malicious actors or cyberattacks.

Table 10 demonstrates that there are two distinct cybersecurity keywords, both of which link to the same domain. One constitutes “cyber security” with 998 occurrences, whereas the other constitutes “cybersecurity” with 945 occurrences, totaling 1943. Machine learning is the second most frequent keyword, with a total of 1006 occurrences. The results also show that machine learning, deep learning, learning systems, learning algorithms, artificial intelligence, decision trees, and neural networks have occurrences of 1006, 722, 592, 402, 267, 246, and 178, respectively. These keywords can be grouped under artificial intelligence, which has a total of 3,413 occurrences. This result demonstrates that researchers are paying more attention to cybersecurity and are presently using AI technologies such as machine learning and deep learning to address cybersecurity problems. Machine learning and deep learning approaches are some of the AI techniques that can be utilized to combat sophisticated attacks and protect cyberspace from cyberattacks. These AI approaches can generate usable and predictive insights that help organizations and cybersecurity practitioners make better decisions and protect cyberspace from sophisticated attacks.

The use of AI algorithms, including machine learning and deep learning, for detecting cybersecurity threats has become increasingly prominent. Machine learning involves training models on past data to classify, detect, or reveal patterns (Bhandari et al. 2023; Oseni et al. 2023). Machine learning algorithms serve as the core structure for numerous AI models in cybersecurity and enable the processing of structured data for cybersecurity threat assessment.

On the other hand, deep learning outperforms conventional machine learning methods by dealing with nonlinear, complex pattern identification (Alkhudaydi et al. 2023). In recent times, deep learning approaches have demonstrated exceptional abilities in identifying com-

Table 10 Top 20 author’s keywords

Words	Occurrences
Machine learning	1006
Cyber security	998
Cybersecurity	945
Network security	886
Intrusion detection	745
Deep learning	722
Learning systems	592
Computer crime	476
Learning algorithms	402
Internet of things	368
Malware	321
Intrusion detection systems	289
Anomaly detection	284
Artificial intelligence	267
Decision trees	246
Classification of information	244
Cyber attacks	206
Crime	187
Neural networks	178
Feature extraction	165

plex patterns in unstructured data, such as network traffic or system logs. This ability proves especially beneficial in the cybersecurity domain, where subtle and evolving threats demand sophisticated detection systems.

Deep learning models are gaining recognition in cybersecurity for their excellent ability to identify, respond to, and thwart various attacks aimed at interconnected assets, such as network infrastructure (Nasir et al. 2022). The integration of machine learning and deep learning algorithms within AI models generates a synergistic approach that leverages their respective strengths to implement effective defense mechanisms against cyber threats (Oseni et al. 2023).

Figure 7 represents the word TreeMap of keywords that sparked researchers' interest in AI cyberattack detection research. Figure 7 depicts that intrusion detection, artificial intelligence, anomaly detection, and the Internet of Things (IoT) are common keywords in cybersecurity. Figure 7 also shows that deep learning, cybersecurity, malware detection, and cyberattacks are common keywords in machine learning. The visualization in Fig. 7 indicates that AI techniques such as machine learning and deep learning have the potential to improve the security of network systems and cyber-physical systems. This also suggests that machine learning and deep learning techniques can enhance intrusion detection systems, anomaly detection infrastructures, malware analysis, and denial-of-service detection systems.

Other than the TreeMap, Fig. 8 depicts a topic dendrogram of keywords related to cybersecurity and machine learning to associate these numerous combination keywords in depth. The hierarchical order and relationship between the keywords, produced by hierarchical clustering, are represented by the topic dendrogram in Fig. 8. Investigating and interpreting the various clusters is made easier by the vertical lines and cuts in the dendrogram. Figure 8 seeks to estimate the approximate number of clusters to ease future research discussions; it does not aim to discover the optimal number of associations between clusters. Other than convolution neural networks, convolution, and convolution neural networks, the remaining elements are the keywords in AI-enabled cybersecurity articles.

The dendrogram illustrates that the classification of information areas represents the different branches of the AI-enabled cyberattack classification model (neural networks, deep learning, long short-term memory, deep neural networks) that can be utilized to classify malicious data or information on network architecture (Li et al. 2022). The security of data

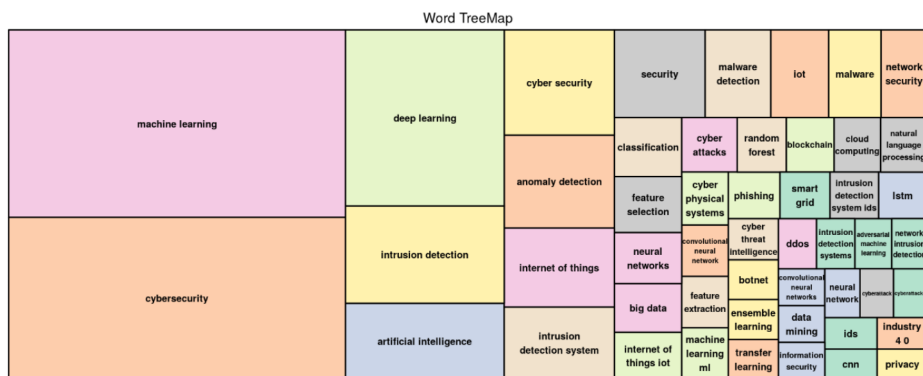


Fig. 7 Word treemap

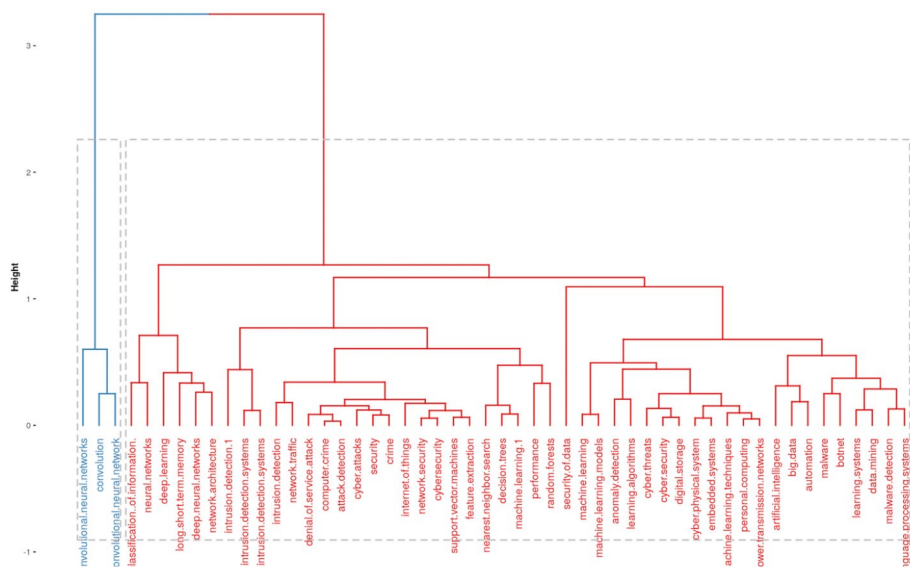


Fig. 8 Topic dendrogram

areas highlights researchers' interest in machine learning models and learning algorithms for detecting anomalous data and executing threat intelligence, while the artificial intelligence area emphasizes researchers' interest in using AI and big data for automation, malware detection, and botnet detection (Rakhsha et al. 2021). This demonstrates that researchers have confidence in AI approaches and have applied them in their work to defend against sophisticated cyberattacks.

The word cloud in Fig. 10 provides a summary of the overall keywords and identifies the words most frequently used by researchers worldwide. The word cloud in Fig. 9 illustrates the terms used in research publications. In Fig. 9, concepts such as computer security, attack detection, crime, cyber threats, embedded systems, network traffic, network design, malware detection, and attack detection performance are highlighted. The word cloud indicates that AI holds significant promise in these research domains and suggests an increase in AI-enabled cybersecurity models in the years ahead.

The search trends based on the examined keywords are shown in Fig. 10. The research started in 2014. Initially, it identified research topics related to information science, network fusion virtualization, mercury (metal), vulnerabilities, heterogeneous data, language models, and data handling. In 2018, the research focused on the development of software-defined networking, traffic analysis, and security analysis systems, computational theory for computer networks, big data, and artificial intelligence. In 2020, the research shifted toward the development of machine learning approaches, data mining, neural networks, and learning systems for data security and continuous authentication.

Between 2021 and 2022, the research emphasized the development of machine learning and deep learning models for intrusion detection, network security, phishing detection, and AI-enabled models for securing the Internet of Things (IoT). This period also focused on



evaluating the performance of AI-enabled cybersecurity models. In 2022, several publications aimed to improve the performance of AI-enabled intrusion detection systems.

Figure 11 illustrates that most institutions have adopted machine learning and deep learning to combat cyberattacks. For example, the University of Murcia and La Trobe University applied machine learning and deep learning to enhance cybersecurity infrastructures and improve anomaly detection. Meanwhile, King Abdulaziz University and Deakin University focused on addressing intrusion detection, anomaly detection, and cybersecurity issues using machine learning and deep learning. Deakin University and Kennesaw State University are also interested in integrating machine learning and deep learning for IoT security.

4.7 Total citations

Table 11 lists the sources or publications that were ranked in the top 20 and frequently cited in other papers. This data reveals that certain papers were often cited during specific years, demonstrating that some articles are particularly relevant within certain periods. The number of citations increases significantly when authors combine AI-enabled cyberattack detection with other fields, especially data mining and big data related to threat intelligence.

The highest number of references has been received by an article published in 2019 (Alom et al. 2019); this article has received the most citations to date. In terms of the number of citations received over multiple years and the ranking attained, one article from 2016 (Nunes et al. 2016), one from 2017 (Wang 2017), seven from 2018 (Biggio et al. 2018; Fernandez Maimo et al. 2018), five from 2019 (Alom et al. 2019; Salah et al. 2019; Alrashdi et al. 2019), five from 2020 (MahdaviFar et al. 2019; Aldweesh et al. 2020; Sarker et al. 2020; Gamage et al. 2020), and one article from 2021 (Sarker and Sarker 2021) were highly significant. This demonstrates that the article's coverage of AI-enabled cyberattack detection approaches is of the highest quality.

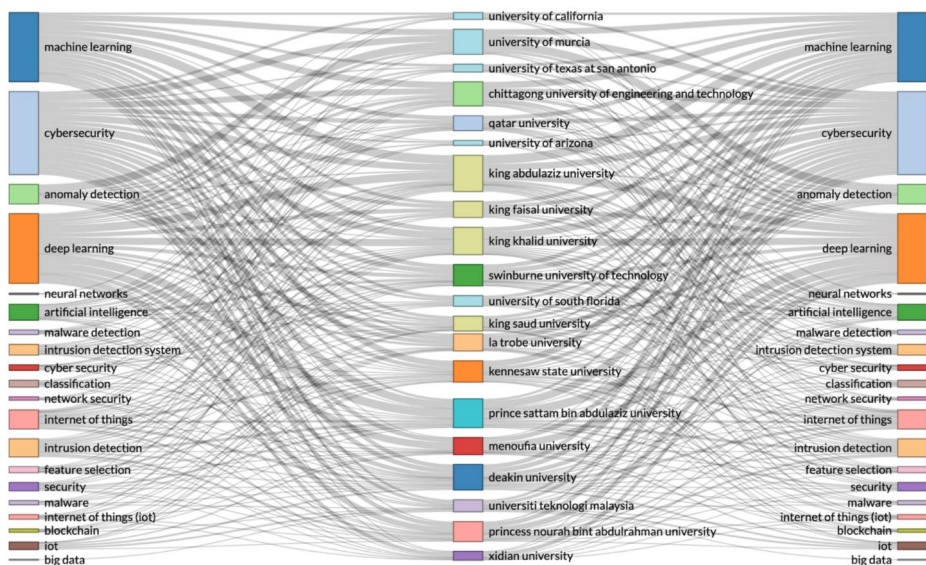


Fig. 11 Multi-field visualization between keywords and affiliation

Table 11 Top 20 cited articles and sources

Paper	Total Citations	TC per Year
(Alom et al. 2019)	585	146.25
(Biggio et al. 2018)	479	95.8
(Xin et al. 2018)	456	91.2
(Diro et al. 2018)	431	86.2
(Salah et al. 2019)	337	84.25
(Hasan et al. 2019)	262	65.5
(Sarker and Sarker 2021)	260	130
(Abeshu et al. 2018)	195	39
(Liu et al. 2018)	187	37.4
(Aldweesh et al. 2020)	177	59
(Vasan et al. 2020)	147	49
(Gupta et al. 2020)	144	48
(Atat 2018)	140	28
(Fernandez Maimo et al. 2018)	138	27.6
(Mahdavifar et al. 2019)	128	32
(Alrashdi et al. 2019)	122	30.5
(Fernández-Caramés et al. 2018)	111	22.2
(Sarker et al. 2020)	106	35.333
(Nishant et al. 2020)	104	34.667
(Nunes et al. 2016)	104	14.857

It is clear from observing the highest source in Table 11; Fig. 12 that IEEE Access consists of five articles. This demonstrates that IEEE Access offers the majority of information on AI-enabled cyberattack approaches, which explains why many authors cite IEEE Access. This resource is a multidisciplinary, open-access (OA), applications-focused, entirely electronic archive journal that regularly presents the findings of original research or development in all of IEEE's study areas. The articles published in this source are of high interest to readers, innovative, and technically sound.

4.8 Country

This section focuses on the countries participating in AI cyberattack detection research. It highlights the author's country as it appears in publications. It contains data on production output, such as the total number of publications, citations, and the country's collaboration networks. The next sections begin with the number of publications based on countries.

4.9 Country total number of articles

Figure 13; Table 12 depict the countries involved in AI cyberattack detection methods. The United States, with a total of 851 articles, is in first place, followed by India (630) and China (356). It is important to note that these three countries are spread across two distinct continents: North and South America are represented by the United States, while Asia is represented by India and China. India has the highest ranking among Asian countries. According to this data, each continent has a leading country in AI cyberattack detection methods.

Figure 13 shows that blue colors are more prevalent in various North and South American countries. This demonstrates that North and South American countries, particularly the

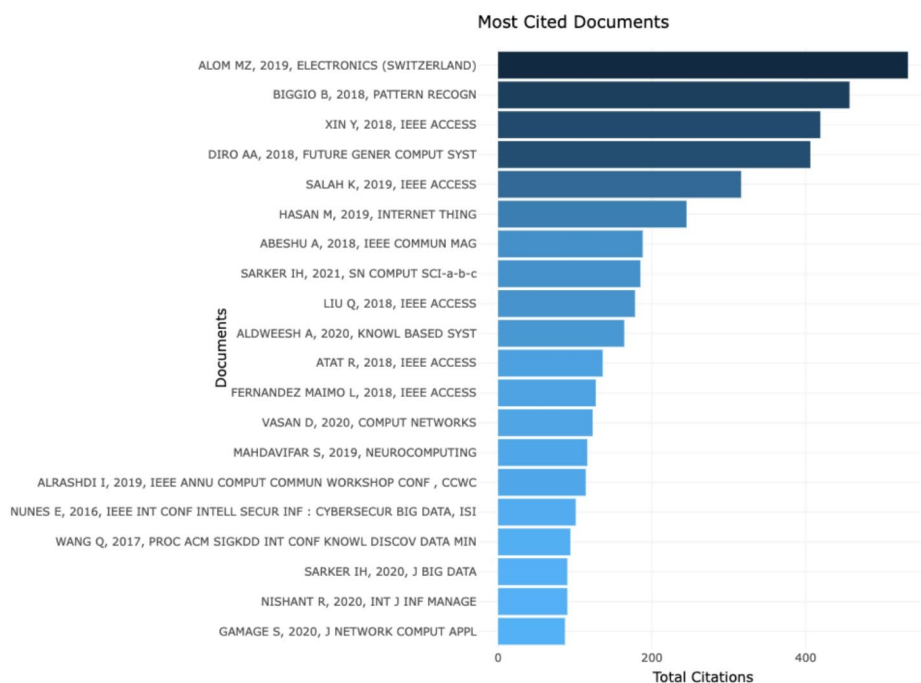


Fig. 12 Most cited sources

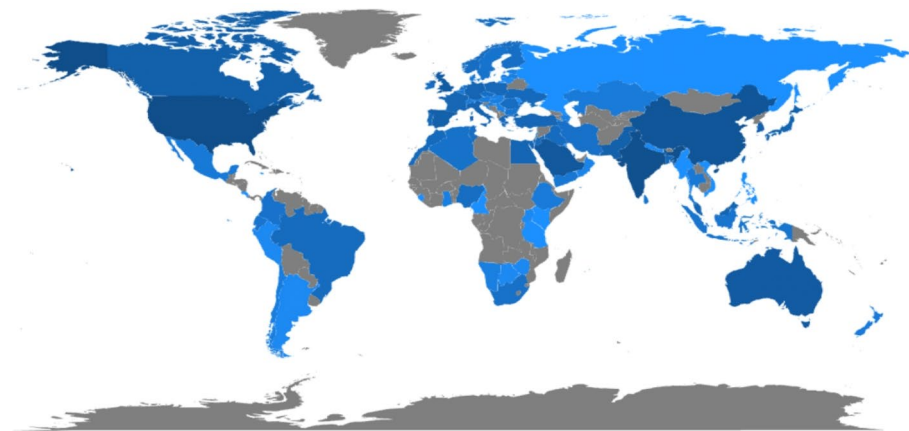


Fig. 13 Country articles production

United States and Canada, have adopted AI cyberattack detection approaches in their studies more frequently than other continents. Additionally, more countries in North and South America are attempting to contribute to AI cyberattack detection methods, as evidenced by the presence of numerous small blue areas encircling the continents. This data indicates that

Table 12 Top 20 ranking countries total number of articles

Country	Frequency
USA	851
INDIA	630
CHINA	356
SAUDI ARABIA	264
AUSTRALIA	202
UK	187
CANADA	123
SPAIN	117
PAKISTAN	114
SOUTH KOREA	88
JAPAN	81
ITALY	77
EGYPT	69
MALAYSIA	69
GERMANY	66
FRANCE	62
BANGLADESH	60
TURKEY	58
GREECE	54
POLAND	49

there are more AI cybersecurity detection researchers in North and South America, and we may expect an increase in publications in the near future.

Asia is also a continent that is actively engaged in AI cybersecurity detection research, in addition to North and South America. India, China, Saudi Arabia, Pakistan, Japan, South Korea, Malaysia, Turkey, and Bangladesh are the countries that lead the Asian continent, as evidenced by the substantial blue colors and the overall quantity of articles. This statistic shows that, in comparison to other nations on this continent, these countries recognized the full potential of the AI cyberattack detection model and contributed to publications with their ideas.

Several factors, including strategic initiatives and investments in AI-driven cybersecurity technologies, can be attributed to the United States' leading position in AI cyberattack detection research, as evidenced by its total of 851 articles. Leading figures in the defense and cybersecurity industries have highlighted the role AI plays in combating cyber threats and see it as essential to the future of cybersecurity (Hoadley and Lucas 2024).

The United States has taken a proactive stance in utilizing AI capabilities for cyber defense, as evidenced by the large number of publications it has produced on the subject. Project Voltron and the Defense Advanced Research Projects Agency (DARPA) Cyber Grand Challenge are examples of initiatives that show the nation's dedication to creating and implementing autonomous cybersecurity systems, adding to the vast amount of research that has been produced in this domain (Hoadley and Lucas 2024). In addition, a strong network of academic institutions, business partnerships, and government assistance helps the U.S. develop its AI cyberattack detection ecosystem and promotes innovation in the research domain (Hoadley and Lucas 2024).

The interdependence of talent, research output, and technological innovation is highlighted by the United States' notable advantage over India and China in AI cyberattack

detection research, with a five-fold talent gap and significant advantages in research and commercial AI (Schmidt and Schmidt 2022). The United States' talent pool, which is well-known for its breadth and depth in AI-related fields, is essential to advancing cutting-edge research and creating innovative commercial AI solutions. The nation's leadership in AI cyberattack detection techniques and broader AI applications is a result of this synergy between elite talent, outstanding research, and commercial viability (Schmidt and Schmidt 2022). As evidenced by its high ranking in the number of publications, citations, and contributions to the global AI cyberattack detection landscape, the United States is recognized as a leader in AI-driven cybersecurity solutions due to this broad strategy, investments in talent development, strategic investments, and a supportive research environment.

There are several factors and policies that have contributed to India's rise to prominence in AI cybersecurity research. The Government of India's Ministry of Commerce and Industry established the AI task force as part of a proactive strategy to prepare for the 4.0 Industrial Revolution and leverage AI to transform the economy. The task force placed priority on AI's applicability across various sectors, including cybersecurity, as evidenced by its identification of ten key domains critical to India's development (SETS 2020).

A survey was conducted in India by the AI task force in 2018. A key factor in identifying the infrastructure, solutions, training, and educational landscape concerning AI and cybersecurity at the national level in India was the objective of the survey. This effort revealed India's concentrated expertise and thriving ecosystem in AI cybersecurity research and development, leading to the identification of specialized research centers, ongoing projects, and industry collaborations (SETS 2020).

Due to its strategic alignment with AI's potential across a range of industries, as well as its focused initiatives, collaborative efforts, and highly qualified workforce, India has established itself as a global leader in AI cybersecurity research and development (SETS 2020). This position allows India to address challenging issues and stimulate innovation and research output in the AI cybersecurity domain.

Several interrelated factors combine to propel China's notable contributions to the field of AI cybersecurity research, making it a prominent player in the field. China's deliberate focus on cybersecurity and AI as essential elements of national growth has prompted large-scale investments, policy frameworks, and programs meant to promote research excellence. This strategic focus has accelerated China's technological advancement and encouraged an environment favorable to innovative and cutting-edge research (Demchak and Demchak 2019).

China is fortunate to have a large pool of highly qualified researchers, engineers, and cybersecurity and AI specialists. China's emphasis on STEM education, skill development, and a culture that values technological innovation has produced this robust workforce. China actively participates in partnerships and international collaborations, which promote knowledge sharing and cooperative research initiatives on a global scale (Demchak and Demchak 2019).

A few key government programs, like the "Next Generation Artificial Intelligence Development Plan" that the State Council of China unveiled on July 20, 2017, have further cemented China's leadership in AI cybersecurity research output (Srivastava and Srivastava 2018). With its strategic goals, investments, and policy frameworks to support research, development, and innovation in AI technologies, including cybersecurity, this all-inclusive plan provides a clear path for China to become a global leader in AI. China's leadership

position in this important field is largely due to the strategic alignment of its research priorities and government initiatives, which strengthen China's capabilities and commitment to advancing AI cybersecurity research globally.

Additionally, financial support and funding from the Chinese government play a vital role in motivating researchers and institutions to concentrate on AI cybersecurity research (Demchak and Demchak 2019; Yang and Yang 2014). China is making a concentrated effort to excel and lead in this important field because its research priorities are in line with its strategic goals and ambitions to establish itself as a global leader in cybersecurity and AI. These interrelated factors promote China's leading position in AI cybersecurity research output, indicating its dedication to advancing knowledge and skills in this progressively important research domain (Demchak and Demchak 2019).

4.10 Corresponding authors country publications

This section addresses the corresponding author's country publications based on single-country publications (SCP) and multiple-country publications (MCP) in AI cyberattack detection. Additionally, it seeks to track the global network of collaboration for publishing research on AI cyberattack detection. Table 13 demonstrates that every country collaborated with three or more other countries. Table 13 shows that the USA, India, China, Saudi Arabia, and Australia are the top five leading countries.

Figure 14 provides a visualization that offers insight into the data, including both SCP and MCP. Figure 14 demonstrates that the USA ranked first in both SCP and MCP, indicating that the USA actively published in both SCP (green colours) and MCP (orange colours). India and China came in second and third, respectively. Table 13 demonstrates that Turkey and Portugal ranked bottom in MCP.

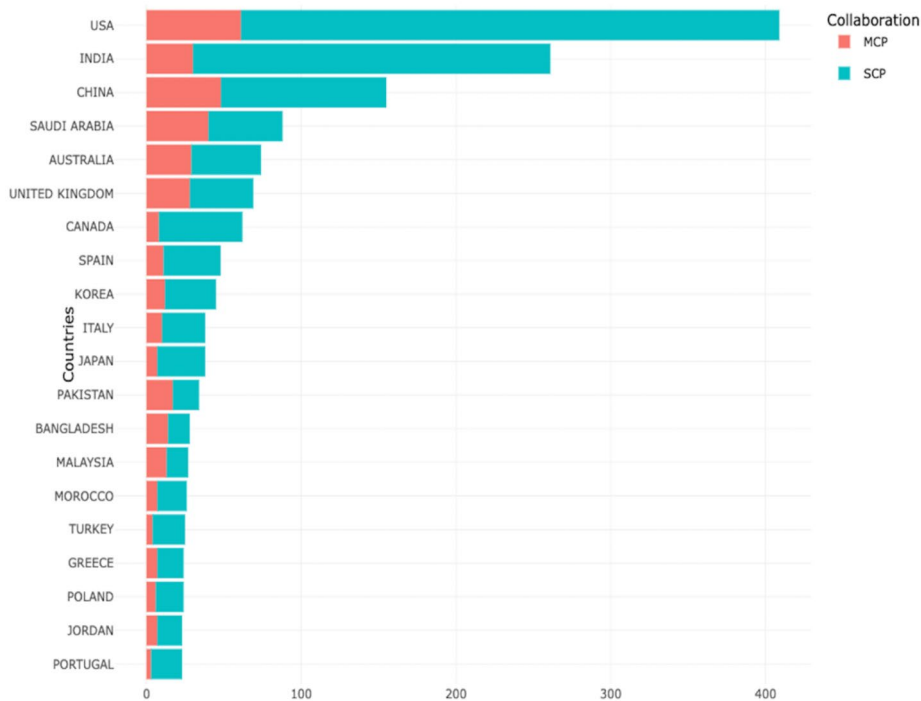
It is noteworthy that the highest MCP values were lower than the SCP, which were 348 and 61, respectively. This pattern demonstrates that most countries are keener to collaborate and publish AI cyberattack detection research in a single country than in several. Nevertheless, Bangladesh and Malaysia expressed an interest in partnering with other countries, as they have 14 and 14, and 14 and 13 papers in the SCP and MCP, respectively. This result shows that Bangladesh and Malaysia are more inclined to collaborate with other countries in AI cyberattack detection research.

4.11 Country collaboration map

Figure 15; Table 14 depict the countries that collaborate globally. Countries that collaborated are shown in blue. The darker blue indicates more frequent global collaboration. The result shows that the USA, Bangladesh, Saudi Arabia, China, and India are among the countries that actively collaborate with other countries. The map shows that the USA is the most active country in terms of collaborating with other countries. The map also depicts that Bangladesh and Saudi Arabia are the second and third most active countries in terms of collaboration. The result indicates that international collaborations can result in more publications than those published in a single country.

Table 13 Top 20 relevant countries by corresponding authors

Country	Articles	Freq	SCP	MCP	MCP_Ratio
USA	409	0.21258	348	61	0.1491
INDIA	261	0.13565	231	30	0.1149
CHINA	155	0.08056	107	48	0.3097
SAUDI ARABIA	88	0.04574	48	40	0.4545
AUSTRALIA	74	0.03846	45	29	0.3919
UNITED KINGDOM	69	0.03586	41	28	0.4058
CANADA	62	0.03222	54	8	0.129
SPAIN	48	0.02495	37	11	0.2292
KOREA	45	0.02339	33	12	0.2667
ITALY	38	0.01975	28	10	0.2632
JAPAN	38	0.01975	31	7	0.1842
PAKISTAN	34	0.01767	17	17	0.5
BANGLADESH	28	0.01455	14	14	0.5
MALAYSIA	27	0.01403	14	13	0.4815
MOROCCO	26	0.01351	19	7	0.2692
TURKEY	25	0.01299	21	4	0.16
GREECE	24	0.01247	17	7	0.2917
POLAND	24	0.01247	18	6	0.25
JORDAN	23	0.01195	16	7	0.3043
PORTUGAL	23	0.01195	20	3	0.1304

**Fig. 14** Relevant countries by corresponding authors

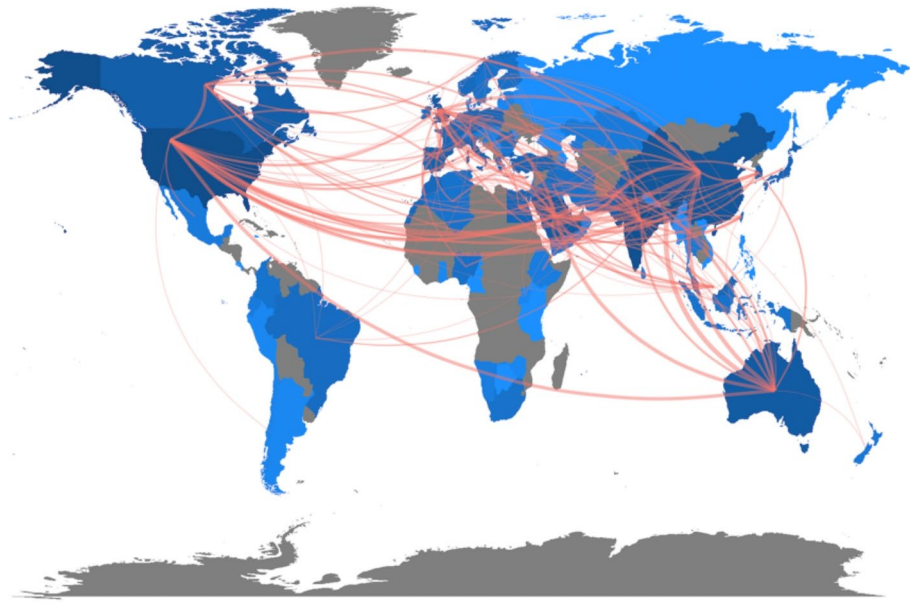


Fig. 15 Collaboration map

Table 14 Country collaboration

From	To	Frequency
USA	CHINA	27
BANGLADESH	AUSTRALIA	20
SAUDI ARABIA	EGYPT	19
CHINA	AUSTRALIA	16
INDIA	SAUDI ARABIA	14
SAUDI ARABIA	PAKISTAN	14
USA	INDIA	14
SAUDI ARABIA	AUSTRALIA	13
CHINA	PAKISTAN	12
CHINA	UNITED KINGDOM	12
USA	AUSTRALIA	12
SAUDI ARABIA	UNITED KINGDOM	11
USA	SAUDI ARABIA	11
UNITED KINGDOM	PAKISTAN	10
USA	CANADA	10
SAUDI ARABIA	TUNISIA	9
UNITED KINGDOM	GREECE	9
AUSTRALIA	UNITED KINGDOM	8
CHINA	CANADA	8
SAUDI ARABIA	JORDAN	8

4.12 Country total citations

This section lists the 20 most influential publications in the world on AI cyberattack detection approaches that have been cited by other researchers. The total and average article citation values are tabulated in Table 15. It provides the total number of citations in order from highest to lowest. The results indicate that the United States ranked first, followed by Australia, China, Italy, and Canada.

Table 13 shows that while some countries received fewer overall citations, their average article citation values were high. These countries include Thailand. This evidence demonstrates that although these three countries published few articles, each one gained several citations worldwide. Additionally, it demonstrates that these countries published high-quality articles on AI cyberattack detection rather than focusing on quantity.

In comparison to Table 12, Italy only produced 77 articles but garnered an impressive 722 citations. Italy tops India, which published 630 more articles but had a lower citation score (553). This is similar to Canada, which produced only 123 articles yet garnered more citations than Saudi Arabia, India, and the United Kingdom. This result indicates that Italy and Canada published in reputable outlets, prompting other researchers to cite and refer to their articles.

The network collaboration between countries, affiliations, and keywords is presented in the subsequent section to shed light on the relationship between these countries.

4.13 Network

The network-based bibliometric analysis of AI cyberattack detection approaches is presented in this section. Both the country collaboration and the network of keyword co-occurrences are included. This network was developed to track the movement of multiple nodes

Table 15 Country total citation

Country	Total citations	Average Article citations
USA	3077	8.384
AUSTRALIA	1327	20.106
CHINA	1273	10.103
ITALY	722	21.235
CANADA	619	10.492
INDIA	553	2.711
SPAIN	462	11.268
BANGLADESH	457	20.773
KOREA	350	9.211
UNITED KINGDOM	230	3.651
SAUDI ARABIA	173	2.37
TURKEY	162	6.75
POLAND	152	7.238
JAPAN	141	4.273
GREECE	139	7.316
BELGIUM	100	12.5
BRAZIL	100	6.667
THAILAND	89	14.833
PAKISTAN	87	3.222
PORTUGAL	80	3.81

that have been interconnected with each other. Figure 15 depicts a record of networking circles of collaboration to highlight countries that are actively interacting. Collaboration is a network that depicts the authors' connections to the network as a result of their co-authorships (Firdaus et al. 2019).

4.14 Country collaboration network

The total number of articles is indicated by the coloured countries in each network node, as illustrated in Fig. 16. The results indicate that the USA, China, Australia, Saudi Arabia, and India are the countries that collaborate the most in publishing articles on AI cyberattack detection approaches. Figure 16 further shows that many lines in the lower portion of the collaboration network are connected. This indicates that, in comparison to the countries in the upper part, the countries in the bottom part were actively collaborating.

The majority of the countries in the lower part are a mix of Asia, Europe, North, and South America. This indicates that countries chose to collaborate when publishing articles in the field of AI cyberattack detection research rather than focusing on a single continent or country.

4.15 Keyword co-occurrences network

All the retrieved articles include keywords following the abstract, as noted in the publications. These keywords are made up of the research areas mentioned in the relevant articles. Figure 17 presents the information needed to determine which fields were involved in AI cyberattack detection research. The keyword co-occurrences of the network are also estab-

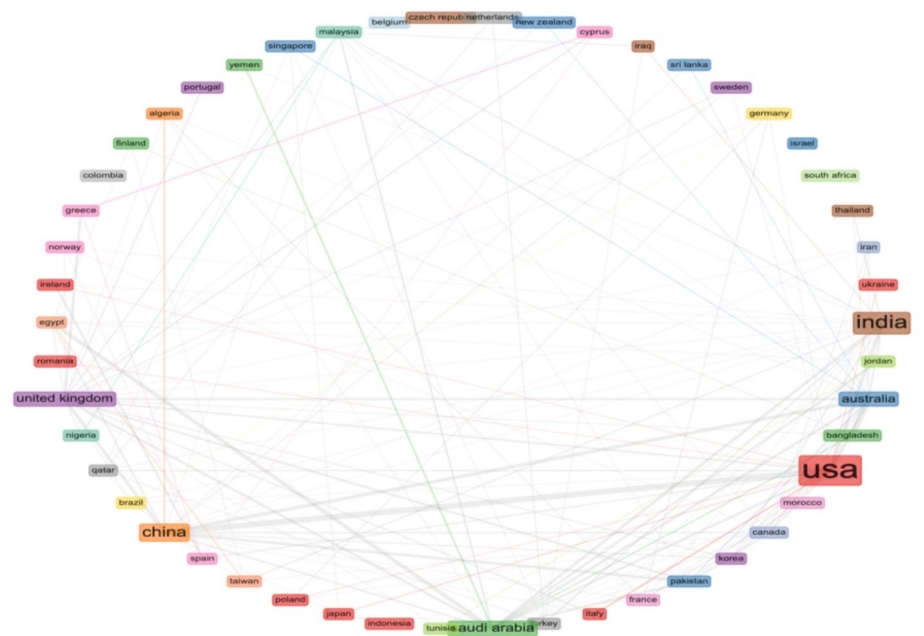


Fig. 16 Country network collaboration map

lished in Fig. 17. The authors use the phrase “machine learning” as a keyword more often than “artificial intelligence,” as shown in Fig. 16. The reason is that machine learning is one of the most widely used artificial intelligence techniques for classifying cyberattacks. There are, therefore, variations in the keyword. However, when it comes to cybersecurity, cyber security, network security, security, security of data intrusion detection, and anomaly detection refer to the same concept.

Furthermore, the most relevant keywords, after cybersecurity, are machine learning, deep learning, learning systems, and learning algorithms. These findings reveal that researchers are currently integrating AI concepts such as machine learning, deep learning, learning systems, and learning algorithms into cybersecurity research as their first attempt. Five different keywords were captured by the keyword co-occurrences network in various capitalizations and semantic formats. They include (1) Intrusion detection, (2) intrusion-detection, (3) intrusion detection systems, (4) network intrusion, and (5) detection systems. The researchers were specifically addressing intrusion detection systems, which is the same domain. To improve intrusion detection systems, AI approaches like machine learning and deep learning are in great demand, as seen in the previous section of this study (4.5 - Authors’ keywords). This finding demonstrates that the trend of integrating AI approaches with intrusion detection systems is excellent, and there will be an increase in the number of publications in these domains shortly.

The analysis also revealed that researchers are interested in using machine learning and deep learning to classify malware and detect malicious network traffic. This demonstrates that AI techniques such as machine learning and deep learning thrive in identifying patterns and extracting behaviours from patterns where humans are rendered ineffective. Machine learning can also detect intrusions based on abnormalities created by malicious or hacked devices, as well as unknown threats, quickly, accurately, and with a low false-positive rate (Navya et al. 2021; Awadallah and Awadallah Awad 2021; Guembe et al. 2022). In addition to the country collaboration map and keyword co-occurrence network, the next section discusses the affiliation collaboration network to identify university collaborations.

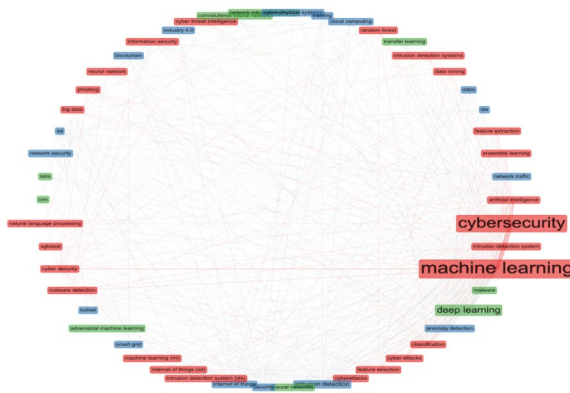


Fig. 17 keyword co-occurrence network

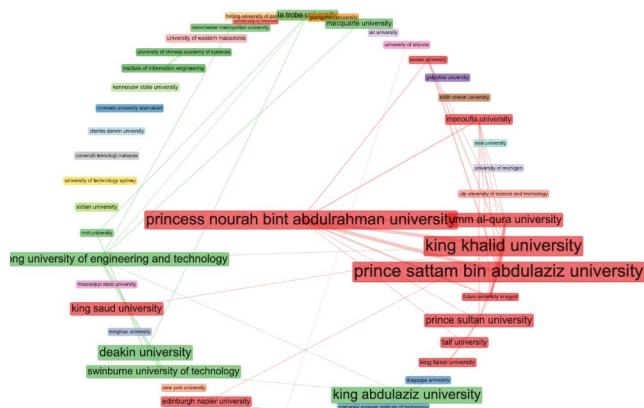


Fig. 18 Affiliation collaboration network

4.16 Affiliation collaboration network

The coloured node in Fig. 18 represents the network collaboration between institutions around the world. The red colour node represents twelve universities that have collaborated between Princess Nourah Bint Abdulrahman University, Umm Al-Qura University, King Khalid University, Prince Sattam Bin Abdulaziz University, Future University in Egypt, Prince Sultan University, Taif University, King Faisal University, Edinburgh Napier University, King Saud University, Sanaa University, and Menoufia University. The green colour node represents eight universities that have collaborated between King Abdulaziz University, Swinburne University of Technology, Deakin University, Chittagong University of Engineering and Technology, RMIT University, Manchester Metropolitan University, La Trobe University, and Macquarie University. The blue colour node at the bottom-right in Fig. 18 represents the collaboration between Alagappa University and Maharaja Agrasen Institute of Technology.

The findings revealed that eight of the universities in the red node are Saudi Arabian universities, while two Egyptian universities (Menoufia University and Future University in Egypt) are Egyptian universities. Yemeni and Scottish universities are represented in the red node by Sanaa University and Edinburgh Napier University, respectively. Figure 18 also revealed that Saudi Arabian universities preferred to team up with universities in Saudi Arabia over others. The reason for this is that when referring to Fig. 7 (Multi-field visualization of keywords and affiliation) and Fig. 18, it is clear that they share a common interest in exploring the potential of machine learning and deep learning in enhancing cybersecurity infrastructures and anomaly detection.

4.17 Statistical test to validate the model

The study utilized the chi-square statistical method to explore the correlation between research output and impact on AI cyberattack detection research. The chi-square test was used to evaluate if there is a significant correlation between two categorical variables. This statistical method evaluates both the quantity and scope of research contributions, assessed by the number of published articles, and evaluates their influence and relevance, often mea-

sured through metrics such as total citations, within the domain of AI cyberattack detection research.

The study achieved a computed chi-square result of 191.8087. This result indicates the strength of the correlation between the total citations and the number of articles across the countries in the dataset. A higher chi-square value indicates a more pronounced correlation between the variables. The p-value associated with the chi-square statistic is roughly $1.439\text{e-}30$, indicating an extremely small value. This small p-value provides compelling evidence against the null hypothesis, indicating that the correlation between total citations and the number of articles is highly unlikely to occur by chance. At a standard significance level of 0.05 in statistical analysis, we reject the null hypothesis (H_0). This implies that we have enough evidence to conclude that there is statistical significance in the correlation between total citations and the number of articles published by each country.

This significant correlation suggests a substantial relationship between the research output (number of articles) and its impact (total citations) for each country. Countries with greater numbers of published articles exhibited higher total citations, indicating a greater level of impact and influence within their respective research domains. Additionally, the significant correlation identified in the chi-square test indicates that countries with robust research outputs in AI cyberattack detection, as measured by both the number of articles and total citations, play a significant role in shaping advancements and practices in global cybersecurity.

This result is valuable to stakeholders in the cybersecurity domain, such as researchers, industry experts, funding bodies, and policymakers. It can be utilized to assess the productivity and influence of research activities in AI cyberattack detection. It also offers insights into the leading countries in terms of both the quantity and quality of research outputs in the research domain.

4.18 Practicability and cost efficiency of the study

The proposed bibliometric analysis model is highly practical in real-time because it integrates up-to-date data from digital libraries and citation indexing services such as Scopus, ensuring timeliness and relevance. Additionally, the integration of automated data collection through APIs will facilitate uninterrupted real-time data acquisition, while robust computational resources support the required data processing, statistical analysis, and visualization. The model also demonstrated a significant correlation between research output and impact within the AI cyberattack detection domain, thus effectively providing valuable and timely insights for industry experts, researchers, policymakers, and funding bodies.

In terms of cost-efficiency, the study focuses on minimizing expenses while maintaining effectiveness by using APIs from digital libraries and citation indexing services for automated data collection, which reduces the need for manual data entry and updates and lowers labour costs. The study also uses open-source tools such as Python and R for statistical analysis and visualization, eliminating the need for expensive proprietary software. Additionally, the model relies on existing computational resources within most academic and research institutions (for example, the bibliometrix library), reducing the need for substantial investment in new infrastructure. Finally, the study can implement real-time dashboards and interactive visualizations using widely available, low-cost software, enabling

stakeholders to access and interpret data without requiring specialized training or expensive tools.

5 Threats to the validity of the study

The use of the Scopus database might undermine the study's validity. For example, coverage bias may result from the Scopus database not including all relevant articles in the research domain. Subjectivity may be introduced by Scopus's indexing criteria and decisions. The accuracy and completeness of the dataset may be impacted by certain articles that are either completely overlooked or indexed differently. Furthermore, there might be an underrepresentation of some journal articles, book chapters, editorials, conference proceedings, and brief surveys. Consequently, this might affect the data's comprehensiveness.

Additionally, biases in the algorithms used by bibliometric tools could include a preference for specific types of publications or citation patterns. However, understanding these biases and their potential influence on outcomes is essential for the accurate interpretation of the findings.

6 Conclusion and future research direction

Machine learning and deep learning have gained significant attention from researchers for their potential in addressing cybersecurity challenges. However, a comprehensive bibliometric analysis in this domain remains limited. This study analyses 2338 articles on AI cyberattack detection published from 2014 to 2024, revealing a preference among researchers for conference proceedings over journals, likely for feedback to enhance later journal submissions. The findings show a rising interest in machine learning and deep learning for detecting cyberattacks, particularly in intrusion detection, network security, phishing detection, and IoT security.

The study further revealed that the United States, India, and China are the top three countries with the most research output on AI cyberattack detection research, while Europe is seeing an increase in researchers in this field. Additionally, Saudi Arabian universities are expected to collaborate, and researchers in Canada and Italy, while publishing less frequently, have gained notable impact for their work.

A statistical analysis was carried out based on chi-square, and the results demonstrated a significant correlation between research output and impact. This means that countries with higher publication volumes receive more citations. The most relevant keywords include network security, federated learning, machine learning, and deep learning, suggesting directions for the development of advanced cybersecurity tools. Given the growing incidence of cybercrime and threats directed toward the Internet of Things (IoT), future research efforts should focus on federated learning and privacy-preserving techniques for enhancing privacy and security on the Internet of Medical Things (IoMT).

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References

- Abeshu A, Chilamkurti N (2018) Deep learning: the frontier for distributed attack detection in Fog-to-Things computing. *IEEE Commun Mag* 56(2):169–175. <https://doi.org/10.1109/mcom.2018.1700332>
- Albahri OS, AlAmoodi AH (2023) Cybersecurity and artificial intelligence applications: a bibliometric analysis based on scopus database. *Mesopotamian J Cybersecur* 2023:158–169. <https://doi.org/10.58496/MJCS/2023/018>
- Aldweesh A, Derhab A, Emam A (2020) Deep learning approaches for anomaly-based intrusion detection systems: A survey, taxonomy, and open issues. *Knowl Based Syst* 189:105124. <https://doi.org/10.1016/j.knsys.2019.105124>
- Alkhudaydi OA, Krichen M, Alghamdi AD (2023) A Deep Learning Methodology for Predicting Cybersecurity Attacks on the Internet of Things, *Information*, vol. 14, no. 10, p. 550. <https://doi.org/10.3390/info14100550>
- Alom M, Taha T, Yakopcic C, Westberg S, Sidike P, Nasrin M, Hasan M, Van BC, Awwal ASS, Asari VK (2019) A State-of-the-art survey on deep learning theory and architectures, *Electronics*, vol. 8, no. 3, p. 292, Mar. <https://doi.org/10.3390/electronics8030292>
- Alshehri A, Khan N, Alowayr A, Alghamdi MY (2022) *Comput Syst Sci Eng* 44(2):1679–1689. <https://doi.org/10.32604/csse.2023.026526>. cyberattack detection framework using machine learning and user behavior analytics,
- Antwarg L, Miller R, Shapira B, Rokach L (2021) Explaining anomalies detected by autoencoders using Shapley additive explanations, expert systems with applications. 186:115736.
- Alrashdi I, Alqazzaz A, Aloufi E, Alharthi R, Zohdy M, Ming H (2019) AD-IoT: Anomaly Detection of IoT Cyberattacks in Smart City Using Machine Learning, in *IEEE 9th Annual Computing and Communication Workshop and Conference (CCWC)*, Jan. 2019, pp. 305–310. <https://doi.org/10.1109/CCWC.2019.8666450>
- Aria M, Cuccurullo C (2017) Bibliometrix: an R-tool for comprehensive science mapping analysis. *J Informetrics* 11(4):959–975. <https://doi.org/10.1016/j.joi.2017.08.007>
- Arpita M, Panchal S (2022) Smart health and cybersecurity in the era of artificial intelligence, in *Information and Communication Technology for Competitive Strategies (ICTCS 2021)*, vol. 401, pp. 41–48. https://doi.org/10.1007/978-981-19-0098-3_5
- Arora P, Jain A (2021) Cybersecurity threats and their solutions through deep learning: A bibliometric analysis. In *Proceedings of the 2021 3rd International Conference on Advances in Computing, Communication Control and Networking (ICAC3N)*, 944– 1949. IEEE. <https://doi.org/10.1109/ICAC3N53548.2021.9725480>

- Atat R, Liu L, Wu J, Li G, Ye C, Yang Y (2018) IEEE Access 6:73603–73636. <https://doi.org/10.1109/access.2018.2878681>. Big Data Meet Cyber-Physical Systems: A Panoramic Survey,
- Awadallah AN (2021) enhancing network intrusion detection model using machine learning algorithms, *Computers, Materials & Continua*, vol. 67, no. 1, pp. 979–990. <https://doi.org/10.32604/cmc.2021.014307>
- Bhandari G, Lyth A, Shalaginov A, Gronli T-M (2023) Distributed deep neural- network-based middleware for cyber-attacks detection in smart iot ecosystem: a novel framework and performance evaluation approach, *Electronics*, vol. 12, no. 2, p. 298. <https://doi.org/10.3390/electronics12020298>
- Biggio B, Roli F (2018) Wild patterns: ten years after the rise of adversarial machine learning. *Pattern Recogn* 84:317–331. <https://doi.org/10.1016/j.patcog.2018.07.023>
- Cojocar I, Cojocar I (2022) A bibliometric analysis of cybersecurity research papers in Eastern Europe: case study from the Republic of Moldova. *Cent East Eur e|Dem e|Gov Days* 151–162. <https://doi.org/10.24989/ocg.v335.12>
- Dayyabu UY, Uppin C, Greenwood J (2020) A bibliometric analysis of cloud security research, *Int. J. Comput. Eng. Technol.*, vol. 11, no. 4, pp. 1–12, Available: <https://www.iaeme.com/ijcet/issues.asp?JType=IJCET&VType=11&IType=4>
- Dehdarirad T, Villarroja A, Barrios M (2015) Research on women in science and higher education: a bibliometric analysis, *Scientometrics*, vol. 103, no. 3, pp. 795–812. <https://doi.org/10.1007/s11192-015-1574-x>
- Demchak CC (2019) China: Determined to dominate cyberspace and AI, *Bulletin of the Atomic Scientists*, vol. 75, no. 3, pp. 99–104. <https://doi.org/10.1080/00963402.2019.1604857>
- Diro A, Chilamkurti N (2018) Distributed attack detection scheme using deep learning approach for internet of things. *Future Generation Comput Syst* 82:761–768. <https://doi.org/10.1016/j.future.2017.08.043>
- Erukala SB, Barthwal A, Kaluri R (2022) Sec-edge: trusted blockchain system for enabling the identification and authentication of edge based 5G networks, *computer communications*. 10–29. Advance online publication 10.1016/j.comcom.2022.12.001
- Fernández-Caramés TM, Fraga-Lamas P (2018) Towards the Internet of smart clothing: a review on iot wearables and garments for creating intelligent connected e-textiles, *Electronics*, vol. 7, no. 12, p. 405. <https://doi.org/10.3390/electronics7120405>
- Fernandez Maimo L, Perales Gomez A, Garcia Clemente F, Gil Perez M, Martinez Perez G (2018) A Self-Adaptive deep Learning-Based system for anomaly detection in 5G networks. *IEEE Access* 6:7700–7712. <https://doi.org/10.1109/access.2018.2803446>
- Firdaus A, Razak M, Feizollah A, Hashem I, Hazim M, Anuar N (2019) The rise of ‘blockchain’: bibliometric analysis of blockchain study, *Scientometrics*, vol. 120, no. 3, pp. 1289–1331. <https://doi.org/10.1007/s11192-019-03170-4>
- Gamage S, Samarabandu J (2020) Deep learning methods in network intrusion detection: A survey and an objective comparison. *J Netw Comput Appl* 169:102767. <https://doi.org/10.1016/j.jnca.2020.102767>
- Gargi S, Saikat G (2022) Cyber security trend analysis using web of science: A bibliometric analysis. *Eur J Mol Clin Med* 7(6):2567–2576
- Guembe B, Azeta A, Misra S, Osamor V, Fernandez-Sanz L, Pospelova V (2022) The emerging threat of AI-driven cyber attacks: a review, *Applied Artificial Intelligence*, pp. 1–34. <https://doi.org/10.1080/08839514.2022.2037254>
- Gupta M, Abdelsalam M, Khorsandroo S, Mittal S (2020) Security and privacy in smart farming: challenges and opportunities. *IEEE Access* 8:34564–34584. <https://doi.org/10.1109/ACCESS.2020.2975142>
- Hasan M, Islam M, Zarif M, Hashem M (2019) Attack and anomaly detection in IoT sensors in IoT sites using machine learning approaches. *Internet Things* 7:100059. <https://doi.org/10.1016/j.iot.2019.100059>
- Hoadley DS, Lucas NJ (2024) Artificial Intelligence and National Security, Congressional Research Service Report, 7-5700 R45178, Version 3, May 8, Available: <https://crsreports.congress.gov/product/details?prodcode=R45178>
- Kaluri R, Rajput DS, Xin Q, Lakshman K, Bhattacharya S, Gadekallu TR, Maddikunta PKR (2021) Roughsets-based approach for predicting battery life in IoT. *Intell Autom Soft Comput* 27(6):3179–3189. <https://doi.org/10.32604/iasc.2021.014369>
- Khaleel YL, Habeeb MA, Albahri AS, Al-Quraishi T, Albahri OS, Alamoodi AH (2024) Network and cyber-security applications of defense in adversarial attacks: A state-of-the-art using machine learning and deep learning methods. *J Intell Syst* 33(1):20240153. <https://doi.org/10.1515/jisys-2024-0153>
- Kumar S, Kumar S (2008) Collaboration in research productivity in oil seed research institutes of India, in *Proc. Fourth Int. Conf. Webometrics, Informetrics and Scientometrics*, vol. 28, Jul
- Li J, Li J, Wang C, Verbeek FJ, Schultz T, Liu H (2024) MS2OD: outlier detection using minimum spanning tree and medoid selection. *Mach Learning: Sci Technol* 5(1). <https://doi.org/10.1088/2632-2153/ad2492>
- Li X, Xiong H, Li X, Wu X, Chen Z, Dou D (2022) InterpretDL: Explaining deep models in paddlepaddle, *J. Mach. Learn. Res.*, vol. 23, no. 197, pp. 1–6, Available: <http://jmlr.org/papers/v23/21-0738.html>

- Liu Q, Li P, Zhao W, Cai W, Yu S, Leung V (2018) A survey on security threats and defensive techniques of machine learning: A data driven view. *IEEE Access* 6:12103–12117. <https://doi.org/10.1109/access.2018.2805680>
- Makawana P, Jhaveri R (2017) A Bibliometric Analysis of Recent Research on Machine Learning for Cyber Security, in *Intelligent Communication and Computational Technologies*, pp. 213–226. https://doi.org/10.1007/978-981-10-5523-2_20
- Mahbooba B, Timilsina M, Sahal R, Serrano M (2021) Explainable artificial intelligence (XAI) to enhance trust management in intrusion detection systems using decision tree model. *Complexity* 2021:1–11. <https://doi.org/10.1155/2021/6634811>
- Mahdavifar S, Ghorbani A (2019) Application of deep learning to cybersecurity: A survey. *Neurocomputing* 347:149–176. <https://doi.org/10.1016/j.neucom.2019.02.056>
- Nakhodchi S, Dehghantanha A (2020) A bibliometric analysis on the application of deep learning in cybersecurity. *Secur Cyber-Physical Systems: Vulnerability Impact* 203–221. https://doi.org/10.1007/978-3-030-45541-5_11
- Nasir M, Javed AR, Tariq MA, Asim M, Baker T (2022) Feature engineering and deep learning-based intrusion detection framework for Securing edge IoT. *J Supercomput* 78:8852–8866. <https://doi.org/10.1007/s11227-021-04250-0>
- Navya V, Adithi J, Rudrawal D, Tailor H, James N (2021) Intrusion Detection System using Deep Neural Networks (DNN), in *International Conference on Advancements in Electrical, Electronics, Communication, Computing and Automation (ICAECA)*, Coimbatore, India. <https://doi.org/10.1109/icaeca5283.8.2021.9675513>
- Nishant R, Kennedy M, Corbett J (2020) Artificial intelligence for sustainability: challenges, opportunities, and a research agenda. *Int J Inf Manag* 53:102104. <https://doi.org/10.1016/j.ijinfomgt.2020.102104>
- Nunes E, Diab A, Gunn A, Marin E, Mishra V, Paliath V, Robertson J, Shakarian J, Thart A, Shakarian P (2016) Darknet and deepnet mining for proactive cybersecurity threat intelligence, in *IEEE Conference on Intelligence and Security Informatics (ISI)*, pp. 7–6. <https://doi.org/10.1109/isi.2016.7745435>
- Oakleaf M (2010) Writing Information Literacy Assessment Plans: A Guide to Best Practice, *CommInfoLit*, vol. 3, no. 2, p. 80. <https://doi.org/10.15760/comminfoLit.2010.3.2.73>
- Ofusori L, Bokaba T, Mhlongo S (2024) Artificial intelligence in cybersecurity: A comprehensive review and future direction. *Appl Artif Intell* 38(1). <https://doi.org/10.1080/08839514.2024.2439609>
- Oseni A, Moustafa N, Creech G, Sohrabi N, Strelzoff A, Tari Z, Linkov I (2023) An explainable deep learning framework for resilient intrusion detection in IoT-Enabled transportation networks. *IEEE Trans Intell Transp Syst* 24(1):1000–1014. <https://doi.org/10.1109/TITS.2022.3188671>
- Pupillo L, Fantin S, Ferreira A, Polito C (2022) Artificial intelligence and cybersecurity, CEPS, Available: <https://www.ceps.eu/ceps-publications/artificial-intelligence-and-cybersecurity-2/>. [Accessed: Feb. 8, 2022]
- Rakhsha A, Radanovic G, Devidze R, Zhu X, Singla A (2021) Policy teaching in reinforcement learning via environment poisoning attacks, *J. Mach. Learn. Res.*, vol. 22, no. 210, pp. 1–45, Available: <https://www.jmlr.org/papers/v22/20-1329.html>
- Razak M, Anuar N, Salleh R, Firdaus A (2016) The rise of ‘malware’: bibliometric analysis of malware study. *J Netw Comput Appl* 75:58–76. <https://doi.org/10.1016/j.jnca.2016.08.022>
- Rahim N Bibliometric Analysis of Cyber Threat and Cyber Attack Literature: Exploring the Higher Education Context, in *Cybersecurity Threats with New Perspectives*, IntechOpen, 2021. Available: <https://doi.org/10.5772/intechopen.98038>
- Salah K, Rehman M, Nizamuddin N, Al-Fuqaha A (2019) Blockchain for AI: review and open research challenges. *IEEE Access* 7:10127–10149. <https://doi.org/10.1109/access.2018.2890507>
- Sarker IH (2021) Machine learning: algorithms, Real-World applications and research directions. *SN Comput Sci* 2(3). <https://doi.org/10.1007/s42979-021-00592-x>
- Sarker IH, Kayes A, Badsha S, Alqahtani H, Watters P, Ng A (2020) Cybersecurity data science: an overview from machine learning perspective. *J Big Data* 7(1). <https://doi.org/10.1186/s40537-020-00318-5>
- Schmidt E (2022) AI, great power competition & national security, *Daedalus*, vol. 151, no. 2, pp. 288–298. https://doi.org/10.1162/daed_a_01916
- SETS, CYBSEC4AI Task force report, society for electronic transactions and security (SETS) under Office of the Principal Scientific Adviser to the Government of India (2020). Accessed: May 9, 2024
- Shevchuk R, Martsenyuk V (2024) Neural networks toward cybersecurity: domain map analysis of state-of-the-art challenges. *IEEE Access* 12:81265–81280. <https://doi.org/10.1109/ACCESS.2024.3411632>
- Srivastava SK (2018) artificial intelligence: way forward for India. *J Inform Syst Technol Manage* 15:1–23. <https://doi.org/10.4301/s1807-1775201815004>
- Vasan D, Alazab M, Wassan S, Naem H, Safaei B, Zheng Q (2020) IMCFN: Image-based malware classification using fine-tuned convolutional neural network architecture. *Comput Netw* 171:107138. <https://doi.org/10.1016/j.comnet.2020.107138>

- Wang Q, Guo W, Zhang K, Ororbia A, Xing X, Liu X, Giles C adversary resistant deep neural networks with an application to malware detection, in Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2017, pp. 1145–1153. <https://doi.org/10.1145/3097983.3098158>
- Wu X, Chen X, Zhan F, Hong S (2015) Global research trends in landslides during 1991– 2014: a bibliometric analysis, *Landslides*, vol. 12, no. 6, pp. 1215–1226. <https://doi.org/10.1007/s10346-015-0624-z>
- Xin Y, Kong L, Liu Z, Chen Y, Li Y, Zhu H, Goa M, Hou H, Wang C (2018) Machine learning and deep learning methods for cybersecurity. *IEEE Access* 6:35365–35381. <https://doi.org/10.1109/access.2018.2836950>
- Yang G (2014) The return of ideology and the future of chinese internet policy, *Critical Studies in Media Communication*, vol. 31, no. 2, pp. 109–113. <https://doi.org/10.1080/15295036.2014.913803>
- Yildiz B, Younes GE (2022) Cyber-Physical systems and cyber security: A bibliometric analysis. *OPUS J Soc Res* 19:1–1. <https://doi.org/10.26466/opusjsr.1063227>
- Yu J, Lu L, Chen Y, Zhu Y, Kong L (2021) An Indirect Eavesdropping Attack of Keystrokes on Touch Screen through Acoustic Sensing, in *IEEE Transactions on Mobile Computing*, vol. 20, no. 2, pp. 337–351, 1 Feb. <https://doi.org/10.1109/TMC.2019.2947468>
- Zahavy T, Ben-Zrihem N, Mannor S (2016) Graying the black box: Understanding DQNs, in Proceedings of the International Conference on Machine Learning, New York, NY, USA, pp. 1899–1908

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